Upper limits, censoring and survival analysis

Eric Feigelson

Center for Astrostatistics, Penn State University
http://astrostatistics.psu.edu

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Outline

1. Concepts for censored data
2. Parametric survival analysis
3. Nonparametric survival analysis
4. Multivariate survival analysis
5. The ASURV code
6. References

Censored data in different fields

Epidemiology Does smoking cause cancer?	Examine longevity of 100 smokers vs. 100 non-smokers	At end of experiment, 80 vs. 65 people are still alive (right-censoring)
Industrial quality control Are widgets Mark 2 more reliable than Mark 1	Test longevity of 100 Mark 2 vs. Mark 1 widgets	At end of experiment, 47 vs. 43 widgets are still working (right-censoring)
Astronomy Do high-redshift starburst galaxies have dust?	Observe 100 $z > 3$ vs. 100 $z < 0.2$ starbursts in mid-infrared	At end of observation, 62 hi-z galaxies are undetected Convert to $L = 4\pi d^2 F$ Compare L_{IR} distributions. (left-censoring)

Left-censoring in astronomy occurs when a previously known object is observed in a new property, but is fainter than some known limit.

Two (redundant) mathematical functions

Survival function:

$$\begin{aligned} S(x) &= Prob(X > x) = \frac{\#obs \geq x}{N} \\ &= 1 - F(x) = 1 - \int_0^x f(s) ds \end{aligned}$$

Hazard function:

$$h(x) = \frac{f(x)}{S(x)} = lim_{\Delta x \to 0} \frac{Prob(x \leq X \leq x + \Delta x | X \geq x)}{\Delta x}$$

$f(x) =$ probability distribution function (pdf)
$F(x) =$ cumulative distribution function (cdf)

Hazard functions are important in human affairs: What is the chance I will die tomorrow? Astronomers generally use survival functions.

Survival analysis for common pdf's

Normal distribution:

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right] \qquad S(x) = 1 - \frac{1}{2}\left[1 + erf\left(\frac{x-\mu}{\sqrt{2}\sigma}\right)\right]$$

Power law (Pareto) distribution:

$$f(x) = \frac{\alpha\lambda^\alpha}{x^{\alpha+1}} \quad S(x) = \frac{\lambda^\alpha}{x^\alpha} \quad h(x) = \frac{\theta}{x}$$

Likelihood for left-censored parametric distribution:

$$\begin{aligned} L &\propto \Pi_{det} f(x_i)\ \Pi_{cens}(1 - S(x_i)) \\ L &= \Pi_{i=1}^n Prob[t_i, \delta_i] = \Pi_{i=1}^n [f(t_i)^{\delta_i}][1 - S(t_i)]^{1-\delta_i}. \end{aligned}$$

Here (t_i, δ_i) where $t_i = min(x_i, c_i)$, $\delta_i = 0$ (if censored) or 1 (if detected), x_i are the detected values, and c_i are the upper limits.

See Cohen (1991) & Lawless (2002) for parametric survival analysis

Kaplan-Meier nonparametric estimator

In 1958, Kaplan & Meier presented the 'product-limit estimator',

$$\widehat{S}_{KM}(x) = \Pi_{x_i \geq x} \left(1 - \frac{d_i}{N_i}\right)$$

where N_i is the number of objects (detected or undetected) $\geq x_i$ and d_i are the number of objects at value x_i. If no ties are present, $d_i = 1$ for all i. It gives the inverse of the e.d.f. with jumps at values where detections occur, but the jumps grow at lower values as the weight of nondetections is redistributed among the detections. For large-N, the asymptotic variance is

$$\widehat{Var}(\widehat{S}_{KM}) = \widehat{S}^2_{KM} \sum_{x_i \geq x} \frac{d_i}{N_i(N_i - d_i)}.$$

The KM estimator has desirable mathematical properties: under random censoring, it is the unique, consistent, asymptotically normal nonparametric maximum likelihood estimator. But it is often not obvious to what degree an astronomical dataset is 'randomly' censored!

Two-sample tests

What is the chance that two censored datasets do not arise from the same underlying distribution? $H_0 : S_1(x) = S_2(x)$

Gehan test (1965) is a generalized Wilcoxon test for survival data. For a left-censored sample x^1 with n objects and x^2 with m objects,

$$\begin{aligned} & = & +1 \text{ if } x_i^1 < x_j^2 \text{ (where } \mathrm{x_i^1} \text{ may be censored)} \\ U_{ij} & = & -1 \text{ if } x_i^1 > x_j^2 \text{ (where } \mathrm{x_j^2} \text{ may be censored)} \\ & = & 0 \text{ if } x_i^1 = x_j^2 \text{ or if the relationship is ill} - \text{determined} \\ W_G & = & \sum_{i=1}^{n} \sum_{j=1}^{m} U_{ij} \qquad \widehat{Var}(W_G) = \frac{mn \sum_{i=1}^{n+m} U_i^2}{(n+m)(n+m-1)}. \end{aligned}$$

Gehan's test statistic W_G is asymptotically normal with zero mean: for $Z = W_G/\sqrt{W_G}$, $P(Z > Z_\alpha | H_0) = \alpha$ where $\alpha = 0.05$.

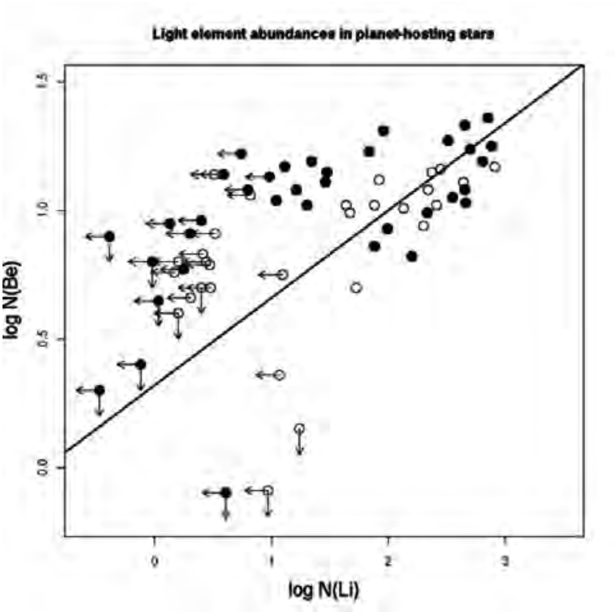
Bivariate correlation

Consider a nonparametric hypothesis test for correlation. Helsel proposes a generalizing Kendall's τ coefficient based on pairwise comparison of data points, (x_i, y_i)–(x_j, y_j).

$$\tau_H = \frac{n_c - n_d}{\sqrt{\left(\frac{n(n-1)}{2} - n_{t,x}\right)\left(\frac{n(n-1)}{2} - n_{t,y}\right)}} \tag{1}$$

where n_c is the number of pairs with a positive slope in the (x, y) diagram, n_d is the number of pairs with negative slopes, $n_{t,x}$ and $n_{t,y}$ are the number of ties or indeterminate relationships in x and y respectively. As the censoring fraction increases, fewer points contribute to the numerator of τ_H, but the denominator measuring the number of effective pairs in the sample also decreases. So τ_H depends on the detailed locations of the censored points.

A doubly-censored bivariate dataset



Linear regression: several approaches

1. Iterative least squares, the familiar linear regression model $y = \alpha + \beta\mathbf{x} + \epsilon$ where $\epsilon = N(0, \sigma^2)$. (Industrial reliability)
2. Accelerated failure-time model $\log y = \alpha + \beta\mathbf{x} + \epsilon$. (Industrial reliability)
3. Tobit regression. (Econometrics)
4. Proportional hazards model (Cox regression) where the hazard rate has an exponential dependence on the covariates, $h(y|\mathbf{x}) = h_0(y)e^{\beta\mathbf{x}}$. MLE estimation and inference. (Biometrics)
5. Buckley-James line permiting non-Gaussian residuals around the line. ϵ is estimated in local regions of x using the Kaplan-Meier estimator. (Biometrics)
6. Akritas-Thiel-Sen line for doubly censored data shown in figure above. (Astronomy)

A confusion in astronomy

Consider the search for relationships between luminosities of objects at different distances observed in two flux-limited surveys. Some researchers advocated that correlations and regression use $F_1 - F_2$ fluxes rather than intrinsic luminosities $L_1 - L_2$ $(L = 4\pi d^2 F)$ because the detections in the $L_1 - L_2$ plane get stretched into a spurious correlation due to multiplication by different distances.

But this problem disappears when the nondetections are included in the $L_1 - L_2$ plane and survival analysis methods are used. Intrinsic relations are generally blurred in the $F_1 - F_2$ plane.

Multivariate survival analysis

There is no multivariate survival analysis!

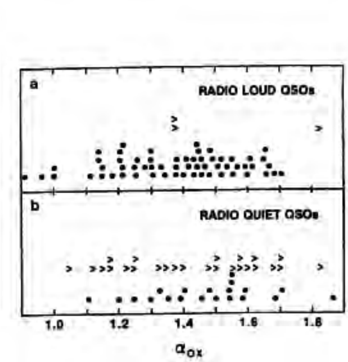
Cox regression is used to find the dependencies of a single censored response variable on many covariates. For example, how does human longevity depend on age, smoking, gender, location, etc.?

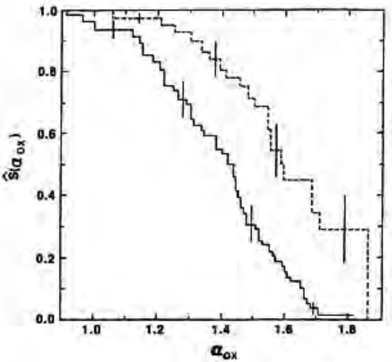
There are no methods that find relationships between p variables, any of which may be subject to censoring. This is the general problem encountered in astronomy. For example, what are the relationships (PCA, clustering, etc) between X-ray, optical, infrared, and radio luminosities in spiral galaxies?

Survival analysis in astronomy

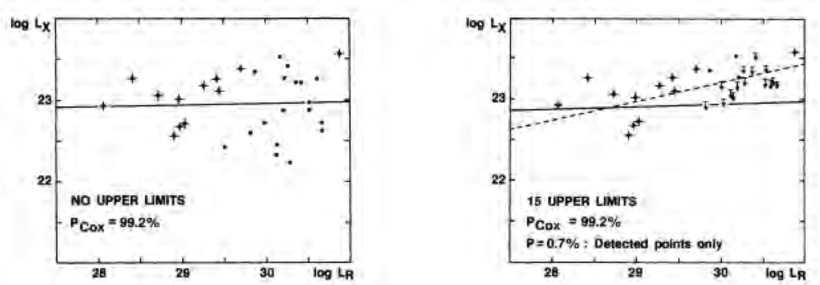
In 1980s, astronomers ‘discovered’ survival analysis and applied them to datasets with upper limits (Feigelson & Nelson 1985, Isobe et al. 1986, Schmitt 1985).

Here are Kaplan-Meier estimators for two right-censored datasets. The KM for heavily censored radio-quiet quasars (dashed line) has big jumps as the limits are redistributed among the detections.





Here is a simulation of uncorrelated X-ray and radio luminosities of a hypothetical sample of galaxies. The left panel has infinite sensitivity in the X-ray band, while the right panel shows finite sensitivity giving upper limits. A spurious correlation is found if only detections are considered (dashed line), but no correlation is found if survival methods (e.g., Cox regression) are used.



The ASURV code

Our group produced a stand-alone Fortran 77 code, Astronomy SURVival analysis (ASURV), which has been widely used. Most of its functionalities are in R/CRAN, and we recommend future work be conducted within R. ASURV implements:

Univariate distribution function
: Kaplan-Meier estimator with confidence limits & quantiles. Assumes random censoring.

Univariate two-sample tests
: Gehan, logrank, and Peto-Prentice tests. Can treat unusual censoring patterns.

Bivariate correlation coefficient
: Generalized Kendall's τ. Can treat censoring in both variables.

Bivariate linear regression
: MLE assuming normal residuals (EM Algorithm), Buckley-James line treating non-normal residuals, Schmitt's binned regression.

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